

AVISHEK ROY

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[LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [TryHackMe](#)

PROFESSIONAL SUMMARY

System-driven Software Engineer with a strong foundation in backend architecture, security, and full-stack development. Actively leveraging AI tools, Agent Skills and Prompt Engineering (via Antigravity) as primary development engines to architect scalable SaaS systems and accelerate the SDLC. Proven track record in optimizing high-performance applications and implementing secure API integrations. Combines rigorous cybersecurity knowledge (CTFs, TryHackMe) with deep systems thinking to perform AI-assisted security audits, correct generated code, and build robust, production-ready software.

PROFESSIONAL EXPERIENCE

SHADHIN MUSIC | Software Engineer | On-site(Bangladesh)

08/2025 – Present

- Architected and engineered a high-performance Next.js application, utilizing AI-assisted workflows to optimize code structure and boost Google Lighthouse scores from 42 to 85.
- Managed the SDLC for organic growth features, driving a large increase in SEO visibility by elevating discoverable pages from 2 to over 500 through dynamic rendering systems.
- Built secure, responsive interfaces and state management systems (Zustand), seamlessly integrating frontend components with backend APIs while ensuring robust access control and data validation.

LYNORG TECHNOLOGIES LIMITED | Software Engineer | On-site (Bangladesh)

01/2025 – 07/2025

- Developed complex, data-driven system interfaces for enterprise ERP solutions using React, TypeScript, and Ant Design.
- Architected real-time data synchronization features using WebSockets, optimizing application performance and component tree rendering for massive datasets.
- Collaborated closely with backend engineers to design, integrate, and test robust REST APIs, ensuring seamless data flow, secure authentication, and overall system stability.

SENSE & RESPONSE SOFTWARE LLC | Software Engineer | Remote (USA)

06/2024 – 12/2024

- Built and maintained scalable React applications with TypeScript, establishing rigorous code standardization and security practices (ESLint) that significantly reduced production bugs.
- Optimized application architecture, loading times, and responsiveness through advanced code-splitting strategies and AI-assisted debugging workflows.

FREELANCE | Frontend Developer (Web) | Remote (USA)

06/2022 – 05/2024

- Managed the complete Software Development Life Cycle (SDLC) for individual clients, translating business requirements into scalable, secure web applications.
- Gained foundational systems experience by designing custom component architectures, implementing secure client-side features, and performing extensive debugging to ensure maintainability.

TECHNICAL SKILLS

AI & Agentic Engineering:

Antigravity, Cursor, Gemini CLI, Claude Code, Agent Skills, MCP Tools, A2A, AI-Assisted Security Audits, Prompt Engineering, AI SDLC Management

Systems Architecture & Integration:

REST API Architecture, Authentication & Authorization (Access Control), WebSockets, System Design, Operating System Concepts (Scheduling etc.), Scalability Validation

Programming Languages:

TypeScript, JavaScript (ES6+), Go (Golang), C++, SQL

Frameworks & Libraries:

Next.js, React.js, Node.js, Redux Toolkit, Zustand, Tailwind CSS, Ant Design, MUI.

Databases, DevOps & Infrastructure:

PostgreSQL, MongoDB, Docker, CI/CD pipelines, AWS (Basic), Linux, Git

Core Engineering & Security:

Data Structures & Algorithms (DSA), Cybersecurity Principles, Performance Optimization, Blockchain Principles

PROJECT HIGHLIGHTS

Shadhin Music – (Performance & SEO Overhaul)

- **Technologies:** Next.js, JavaScript, Tailwind CSS, Zustand
- Engineered a performance and SEO overhaul, boosting the Lighthouse score from 42 to 85 and increasing discoverable pages from 1 to over 500.

All Things API – (Codebase Modernization)

- **Technologies:** React, TypeScript, Material UI, Redux, ESLint
- Modernized a legacy codebase by implementing code standardization to reduce bugs, building a reusable component library, and adding internationalization support.

UI for Lynorg Workflow – (Real-Time ERP Interface)

- **Technologies:** React, JavaScript, Ant Design, Material UI, Redux, WebSockets
- Built a complex, real-time ERP interface with WebSockets, utilizing advanced optimizations like debouncing and code-splitting to ensure high performance.

EDUCATION

B.Sc in Computer Science and Engineering

2019 - 2024

- Khulna University of Engineering & Technology, Khulna, Bangladesh (KUET)

Higher Secondary Certificate (Science)

2016 – 2018

- Notre Dame College, Dhaka.

PUBLICATIONS

First Author: "[A Scalable Cross-Border Payment System based on Consortium Blockchain Ensuring Auditability.](#)"

Contributing Author: Contributed to two additional research papers on blockchain and educational technology, serving as Third Author on "[Sociala](#)" and Fourth Author on "[QEdu](#)".

ACHIEVEMENTS & TECHNICAL GROWTH

Algorithm & Data Structures: Solved over 130 problems on LeetCode ([x64bit](#)), demonstrating a strong foundation in DSA and algorithmic thinking.

Cybersecurity & Problem Solving: Completed 28+ rooms on TryHackMe ([64bitX](#)), gaining practical insights into cybersecurity principles. Ranked 48th in the Intra-University Capture the Flag, showcasing rapid debugging and security skills.

Analytical Skills: Achieved 9th rank in the Intra-KUET Math Olympiad 2019, reflecting strong analytical and problem-solving capabilities.